IALA | AIS Data

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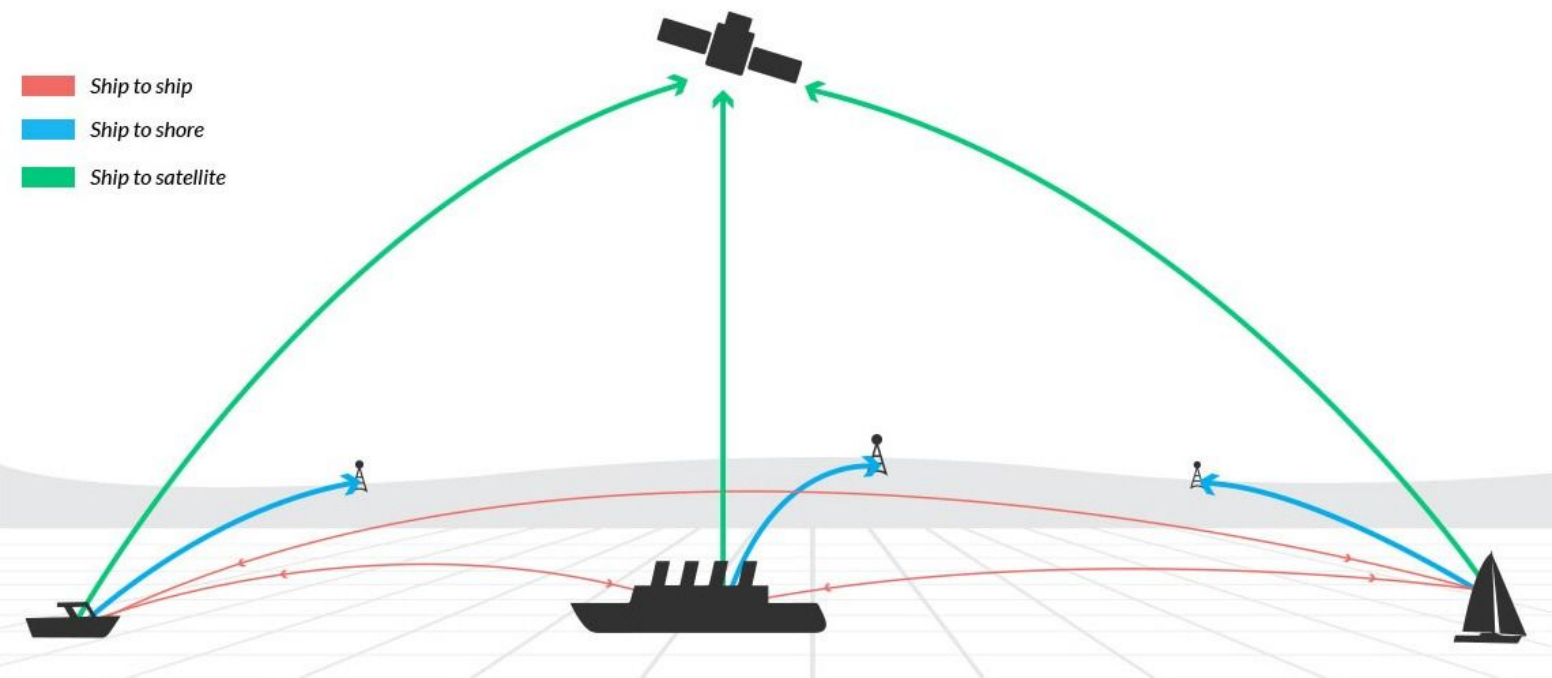
Introduction

- **AIS** stands for **Automatic Identification System**, a technology used by ships to broadcast information about their position, speed, heading, and other data to other ships and to shore-based stations.
- It has revolutionized the way ships navigate, communicate, and operate, leading to improved safety, efficiency, and security in the world's oceans.
- It has become an indispensable tool for vessel tracking and monitoring, maritime safety and security, environmental protection, traffic management, and commercial analysis.

Types of information provided by AIS data

- Vessel identity
- Position and speed
- Navigation status
- Course and heading
- Cargo information

Collection and transmission of AIS data



Applications of AIS data

- Vessel tracking and monitoring:
- Maritime safety and security
- Environmental monitoring and protection
- Traffic management and planning
- Commercial and economic analysis

Our Dataset

```
\s:Cape Schanck A,c:1664582400,T:2022-10-01 00.00.00*7B\!ABVDM,1,1,5,B,17OgKC0P00bF?LQb6fPbrOwp2d0O,0*48
\s:Cape Schanck A,c:1664582400,T:2022-10-01 00.00.00*7B\!ABVDM,1,1,4,B,17Oj?9?00O:F:E?b6<49knwn0<0Q,0*67
\s:Cape Schanck A,c:1664582401,T:2022-10-01 00.00.01*7B\!ABVDM,1,1,7,B,17OkdA?P@p:DorMb;:P@>P@20<0N,0*72
\s:Cape Schanck A,c:1664582401,T:2022-10-01 00.00.01*7B\!ABVDM,1,1,9,A,19NSBB@21h:EIsAajHqS>Rcn0D0;,0*51
\s:Cape Schanck B,c:1664582402,T:2022-10-01 00.00.02*78\!ABVDM,1,1,0,B,17P0?e0001:F@VAb6bu1B`<6081o,0*04
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\s:Cape Schanck A,c:1664582403,T:2022-10-01 00.00.03*7B\!ABVDM,1,1,6,B,17Of=60Oww:F?Job6fRN44ap0h2L,0*3C
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\s:Cape Schanck A,c:1664582406,T:2022-10-01 00.00.06*7B\!ABVDM,1,1,8,B,19NSTE@wj7:G;Aub8n:=Cbd<0<1@,0*3D
\s:Cape Liptrap,c:1664582406,T:2022-10-01 00.00.06*07\!ABVDM,1,1,4,A,19N`2JP00E:HeqganJk0FAJ80D0H,0*34
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```
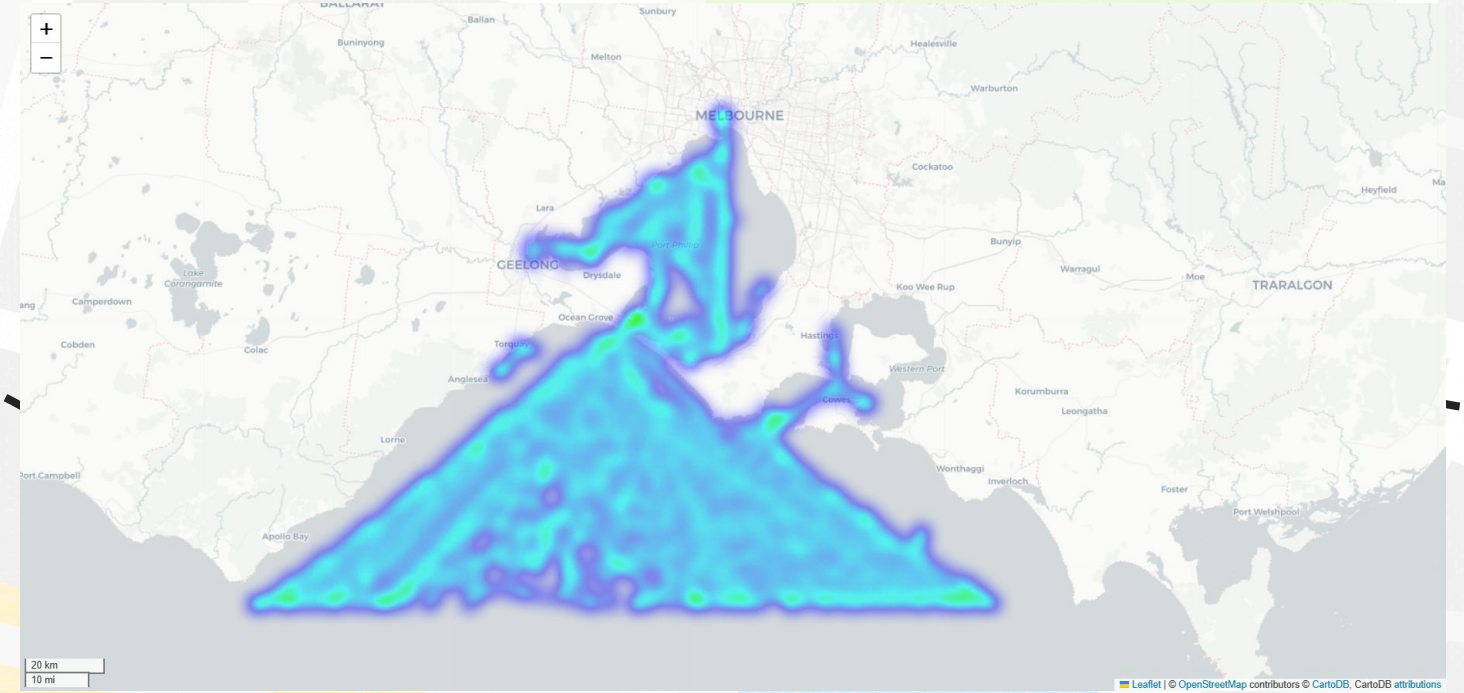

Our Dataset

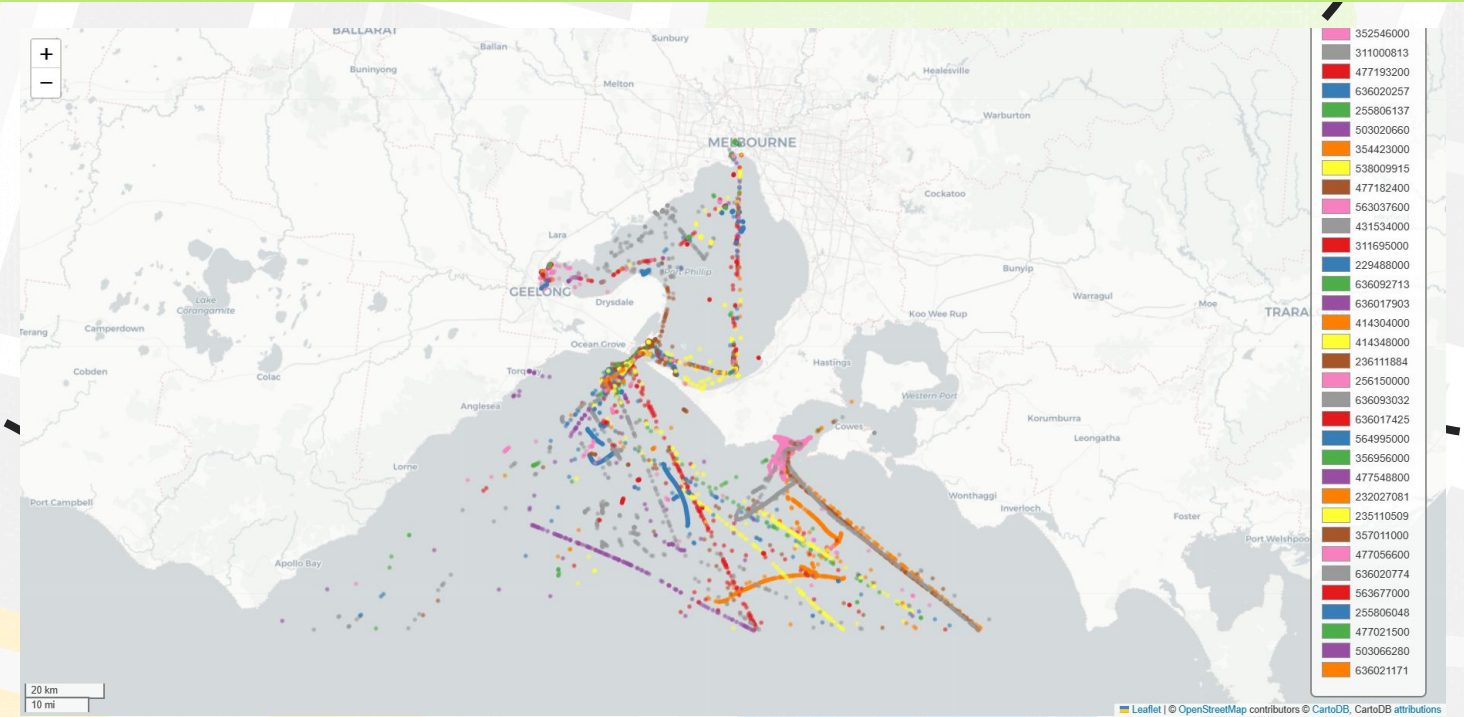
	mmsi	speed	lon	lat	course	heading	coordinates	timestamp
0	503044940	0.0	144.668827	-38.264103	279.3	511	(-38.2641,144.6688)	2022-10-01 00:00:00
1	503090980	3.1	144.651372	-38.278800	251.1	223	(-38.2788,144.6514)	2022-10-01 00:00:00
2	503114820	5.6	144.370050	-38.142932	5.8	8	(-38.1429,144.37)	2022-10-01 00:00:01
3	636015177	11.2	144.486147	-38.819457	82.6	85	(-38.8195,144.4861)	2022-10-01 00:00:01
4	503320500	0.1	144.672760	-38.265620	33.0	262	(-38.2656,144.6728)	2022-10-01 00:00:02
...	...	...	...	...	...	...	...	...
2228740	503066280	2.6	144.386420	-38.341000	121.2	511	(-38.341,144.3864)	2022-10-14 00:16:16
2228741	503382000	0.1	144.364467	-38.113100	67.0	136	(-38.1131,144.3645)	2022-10-14 00:16:16
2228742	503599200	4.0	144.802513	-38.307200	211.3	511	(-38.3072,144.8025)	2022-10-14 00:16:16
2228743	503320500	0.1	144.746167	-38.336167	140.0	99	(-38.3362,144.7462)	2022-10-14 00:16:17
2228744	563677000	16.0	144.818333	-38.320467	104.0	105	(-38.3205,144.8183)	2022-10-14 00:16:17

2228745 rows × 8 columns

Our Experiments: Density of Trajectories



Our Experiments: Stay Point Detection



Our Experiments: Finding GPS Jumps based on Distance

	id	datetime	lat	lon	dist_to_prev	dist_to_next	dist_prev_to_next	gps_jump
2005091	566961000	2022-10-13 23:00:57	-38.745498	145.045917	45.308172	49.654912	94.961703	False
2005092	566961000	2022-10-13 23:03:40	-38.745267	145.045427	49.654912	50.062494	99.712232	False
2005093	566961000	2022-10-13 23:07:41	-38.745042	145.044927	50.062494	35969.301194	35919.239230	True
2005094	566961000	2022-10-13 23:09:03	-38.907440	145.404032	35969.301194	46.781890	35922.519702	True
2005095	566961000	2022-10-13 23:09:10	-38.907228	145.403565	46.781890	55.488590	102.270335	False
2005096	566961000	2022-10-13 23:09:18	-38.906978	145.403010	55.488590	64.773879	120.262389	False
2005097	566961000	2022-10-13 23:09:27	-38.906685	145.402363	64.773879	46.429420	111.202355	False
2005098	566961000	2022-10-13 23:09:35	-38.906478	145.401897	46.429420	55.264520	101.693055	False
2005099	566961000	2022-10-13 23:09:40	-38.906228	145.401345	55.264520	64.288601	119.553119	False

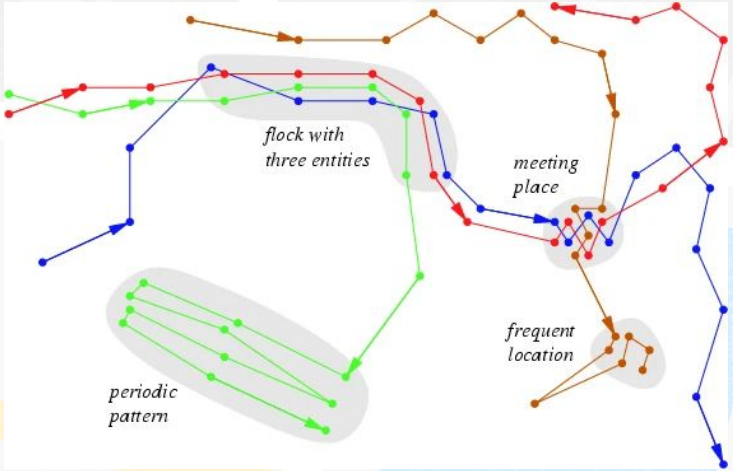
Our Experiments: Prefixspan

	mmsi	coordinates
0	209083000	(-38.3056,144.7546);(-38.3173,144.5939)
1	210415000	(-38.9074,145.6628);(-38.9072,145.6622);(-38.9...
2	212814000	(-38.9074,145.5076);(-38.9071,145.5071);(-38.9...
3	215894000	(-38.9074,145.1972);(-38.9073,145.1968);(-38.9...
4	220216000	(-38.9075,145.5677);(-38.9073,145.5673);(-38.9...

```
[(2, ['(144.386625,-38.090273)']),
 (2, ['(144.386625,-38.090273)', '(144.386638,-38.09027)']),
 (2, ['(144.386625,-38.090273)', '(144.386657,-38.090292)']),
 (2, ['(144.386657,-38.090285)']),
 (2, ['(144.386657,-38.090285)', '(144.386657,-38.090292)'])]
```

Next Steps

- Apply algorithms to detect near misses
- Evaluate the quality of the data
- Apply pattern mining algorithms



Thank You

Any Questions?

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Resources

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